

First Draft of the Capstone Project Proposal

Description & Problem Statement

At my current employer—a fully integrated steel mill—I manage large volumes of rental data from approximately 25 third-party vendors. The current workflow ([Figure 1](#)) is "personality-dependent," relying on manual scrutiny, email collection, and a specific C# upload script. This process is prone to failure if not actively managed because it requires constant "babysitting" to address inconsistent vendor replies and "unclean" data.

The core problem is the administrative burden of manually scrubbing data and hunting down reports to meet tight weekly deadlines. This project aims to automate the intake pipeline, shifting the burden of data integrity back to the vendors, and evolving the system from reactive reporting into a Prescriptive Analytics System.

Three Curricular Areas

- **Software Systems (SS):** To maintain compliance with my company's sanctioned Microsoft environment, the primary interface I will attempt to utilize will be a SharePoint/MS Forms portal. If corporate firewall restrictions prevent vendor access, my backup plan for the project will be to deploy a secondary Django/Python web application featuring secure logins and 2FA security features to act as the "drop zone" for clean data. Both methods will utilize "hard" validation rules to reject improperly formatted data at the source. SharePoint/MS Forms portal. If corporate firewall restrictions prevent vendor access, my backup plan will be to deploy a secondary Django/Python web application to act as the "drop zone" for clean data. Both methods will utilize "hard" validation rules to reject improperly formatted data at the source.
- **Data Management (DM):** I will construct a structured MS SQL Server backend to serve as the "skeleton" of the project. This will include a "Market Reference" table for purchase prices and a consolidated "Historical Rental" table to enable long-term trend analysis. This architectural shift ensures that the data upload fails if the inputs are not accurate, consistent, and complete.
- **Business Analytics (BA):** The "brain" of the project will be an enhanced Power BI Dashboard. Beyond displaying current on-rent assets, it will provide Prescriptive Alerts for "Rent vs. Buy" recommendations by cross-referencing vendor data with estimated market prices.

Final Paper Integration (All Curricular Areas)

The final synthesis paper will incorporate the fourth pillar:

- **Cybersecurity and Networking (CN):** I will articulate the security protocols required to provide vendors with a safe place to enter data without allowing malicious actors to breach the mill's internal network. This includes outlining Multi-Factor Authentication (MFA) and Zero Trust principles to protect the data path while maintaining operational speed.

Project Requirements & Execution

- **GitHub Repository:** All automation logic, Django web code, and SQL schemas will be pushed to the WTAMU-CIDM6395 repository.
- **Functional Artifact:** The prototype will prove the end-to-end flow from secure vendor submission to automated validation and final prescriptive dashboard insight.

Figure 1

